1 Important Constants & Equations

One of the key laws you will use in circuit analysis is **Kirchhoff's Loop Law**. It essentially states that a closed loop forms a voltage difference of zero:

$$\Delta V_{\text{loop}} = \sum_i (\Delta V)_i = 0 \qquad \text{(Kirchhoff's Loop Law)}$$

Another key law you will use in circuit analysis is **Kirchhoff's Junction Law**. It essentially states that the net current into a junction is always zero:

$$\sum I_{\text{in}} = \sum I_{\text{out}} \qquad \text{(Kirchhoff's Junction Law)}$$

In a circuit with linear elements (which are the only elements you will use in this course) the voltage and current are directly proportional:

$$I = \frac{\Delta V}{R} \qquad \text{(Ohm's Law)}$$

The power dissipated by a resistor can be calculated in many different ways:

$$P_R = I\Delta V_R = I^2 R = \frac{\Delta {V_R}^2}{R} \qquad \text{(Power dissipated by a Resistor)}$$

Resistors in series and parallel behave in simple ways:

$$R_{eq} = R_1 + R_2 + \cdots + R_N \qquad \text{(Equivalent Resistance of Series Resistors)}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N} \qquad \text{(Equivalent Resistance of Parallel Resistors)}$$

RC, or Resistor-Capacitor, circuits are crucial to resisting random fluctuations in voltage. The key component is the capacitor, which has a through-current defined by:

$$I_C(t) = C\frac{dV_C(t)}{dt} \qquad \text{(The Current Through a Capacitor)}$$

Applying the above relation, along with fundamental circuit laws, gives us important expressions for the voltage and current over time (Note the emphasis on 'Discharging'!):

$$I = I_0 e^{-t/\tau} \qquad \text{(The Current in an RC Circuit)}$$

$$\Delta V_C = \Delta V_0 (1 - e^{-t/\tau}) \qquad \text{(The Voltage Across a Capacitor in a \textbf{Discharging} RC Circuit)}$$

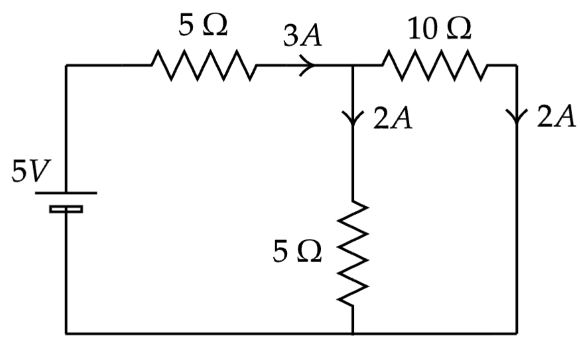
The following constants/relations are important for the study of electricity:

$e = 1.60 \times 10^{-19}$ C (Fundamental Unit of Charge) $m_p = 1.67 \times 10^{-27}$ kg (Mass of Proton)
$m_e = 9.11 \times 10^{-31}$ kg (Mass of Electron) $K = 8.99 \times 10^9$ Nm^2/C^2 (Electrostatic Constant)
$\epsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2 (Permittivity Constant) $\lambda = {}^{dq}/_{dl}$ $\eta = {}^{dq}/_{dA}$ $\rho = {}^{dq}/_{dV}$ $I = {}^{dQ}/_{dt}$ $\tau = RC$

There's not much empty space this time :(

2 Problems

1. Your professor gives you a circuit to analyze:



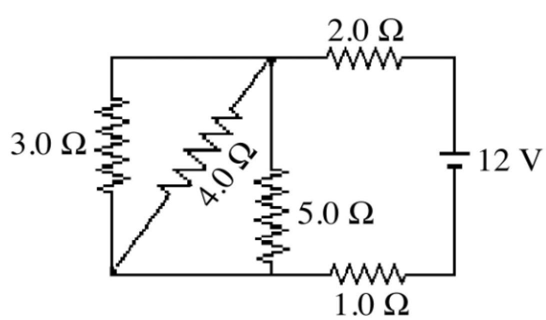
(a) You look at the circuit and begin to wonder if the currents provided are physically possible. Are the currents probable or not? If not, use any of the circuit laws you have learned thus far to explain why it cannot be possible.

(b) Your professor agrees and asks you to scrap the currents and simply reduce the circuit to a single battery and a resistor.

(c) Using the reduced equivalent circuit from part (b), determine the current that must be flowing through the resistor.

(d) **(BONUS).** Using the current you found in part (c), what must be the current through the 10 Ω resistor?

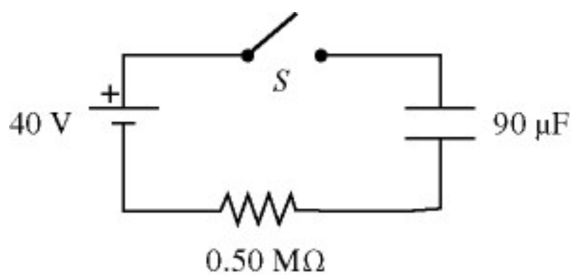
2. Below is a circuit with multiple resistors connected to a $12V$ battery. **(Adapted from Ch. 28 Review #17).**



(a) What is the current in the 3.0 Ω resistor? What is the voltage drop across it?

(b) What is the current in the 4.0 Ω resistor? What is the voltage drop across it?

3. Below is an RC circuit with a switch S. **(Adapted from Ch. 28 Review #31).**



(a) If the switch S is closed a time $t = 0$, which is the current I_0 at time t? What about the current I_∞ as $t \to \infty$?

$I_0 =$

$I_\infty =$

(b) What is the time constant τ of the capacitor in this circuit?

$\tau =$

(c) After closing the switch, how long does it take for the energy stored in the capacitor to equal 50.2 mJ?

4. **(CHALLENGE PROBLEM).** You are designing a simple remote controlled car. The car works by driving a motor with current. The higher the current, the faster the car moves. The circuit you design consists of a single loop containing a resistor and a battery.

   (a) What are two ways you can vary the speed of the car, or in other words, vary the current flowing through the circuit?

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   (b) You decide to use a variable resistor (which is called a *potentiometer*) that can take on values of resistance from 10 to 2000 Ω.

      i. It turns out that the velocity v of your car is related to the current I provided by the circuit using the expression $v = 5\sqrt{I}$.

      Find the velocity in terms of the battery voltage ε and the resistance R.

      ii. You decide to use a AAA battery (which has a voltage difference of $1.5V$) for the car. What is the maximum possible speed your car can reach?

      iii. A typical AAA battery has an internal resistance of about 0.2 Ω. Does this significantly impact the maximum speed of the car?

   (c) **(CHALLENGE BONUS).** The variable resistor is changing its resistance at a rate $\frac{dR}{dt} = 5\ \Omega/s$.

      i. At what rate is the current changing?

ii. Assume that the resistance at time t is $R(t) = 300\ \Omega$. Find the acceleration of the car at this time t.

iii. Assuming that the initial resistance $R_0 = 10\ \Omega$, how much power is expended to reach $R_f = 300\ \Omega$?